

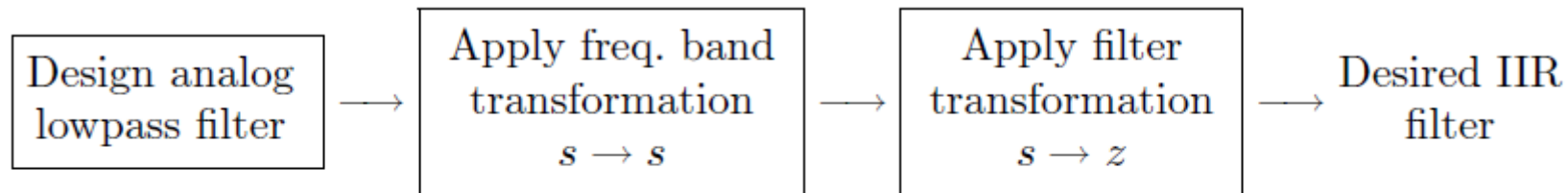
PROJETO DE FILTROS IIR

INTRODUÇÃO

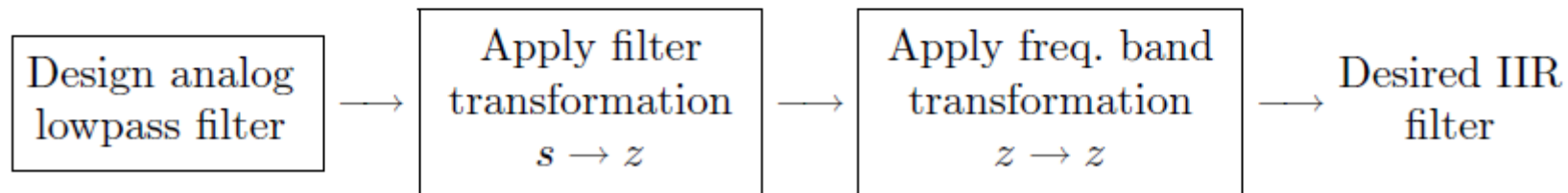
- Filtros IIR tem resposta ao impulso com duração infinita, então eles podem ser relacionados a filtros analógicos.
- Portanto a técnica básica para o projeto de filtro IIR transforma bem conhecidos filtros analógicos em filtros digitais usando mapeamento de valor.
- A vantagem desta técnica reside no fato de que ambos a tabela do projeto de filtros analógicos e o mapeamento estão disponíveis extensivamente na literatura.
- Entretanto, a tabela do projeto de filtros analógicos estão disponíveis somente de filtros passa-baixa.
- Para o projeto dos outros filtros, (FPA, FPF, FCF), aplica-se transformação de banda de frequência para filtros passa-baixa. Estas transformações são também mapeamento de valor complexo e estão disponíveis na literatura, também.

INTRODUÇÃO

Approach 1:



Approach 2:



- Aproximação 2 envolve transformação em frequência no domínio digital.
- Para projetar filtros IIR, vamos:
 - 1) Projetar FPB analógicos;
 - 2) Aplicar transformações no filtro para obter FPB digitais;
 - 3) Aplicar transformações de frequência nas bandas para obter outros filtros digitais a partir do FPB.
- O principal problema dessas técnicas é que não temos controle sobre a fase do filtro. Assim, os projetos de filtros IIR serão apenas em magnitude.

ALGUMAS PRELIMINARES

- Escala Relativa

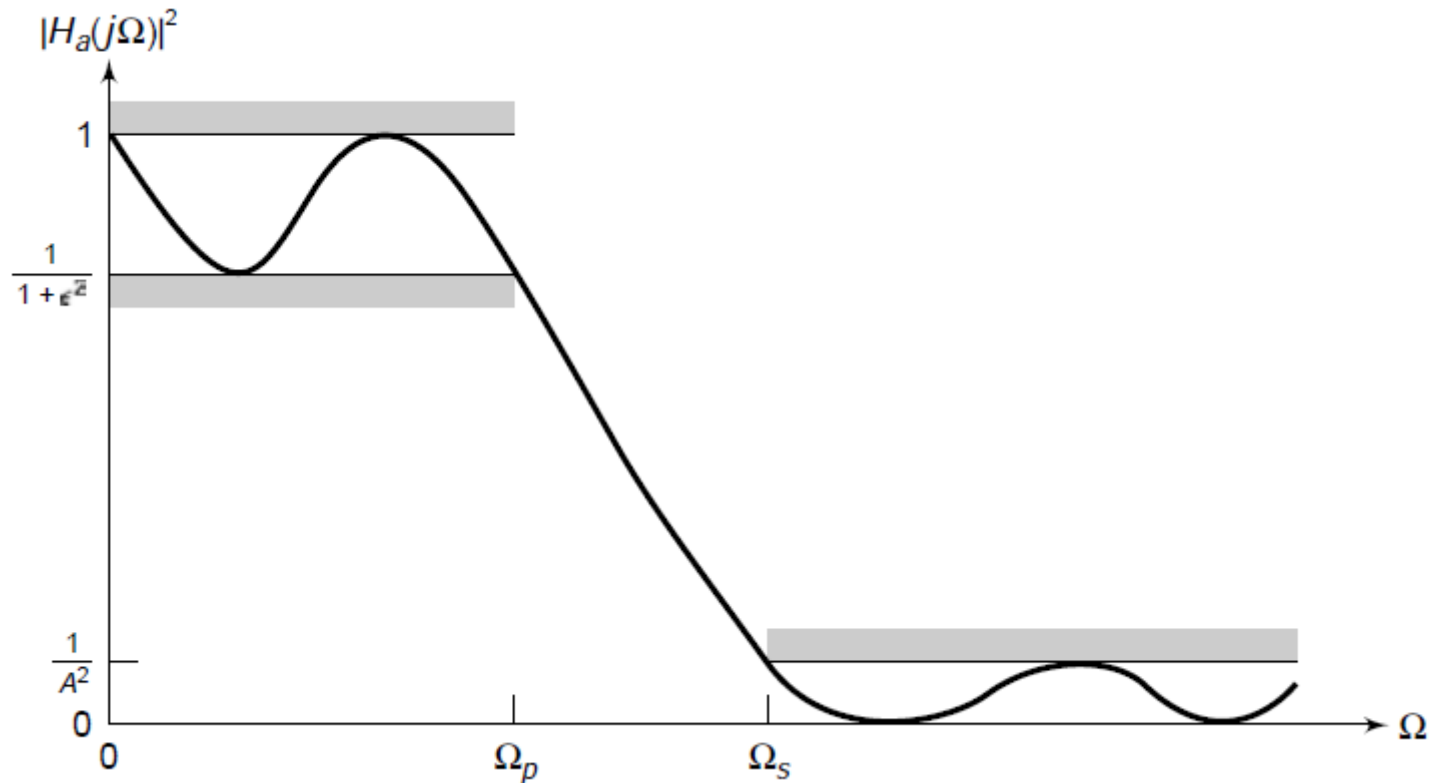
- Seja $H_a(j)$ a resposta em frequência do filtro analógico
- Então as especificações do FPB quanto à resposta quadrática de magnitude são dadas por:

$$\frac{1}{1 + \epsilon^2} \leq |H_a(j\Omega)|^2 \leq 1, \quad |\Omega| \leq \Omega_p$$
$$0 \leq |H_a(j\Omega)|^2 \leq \frac{1}{A^2}, \quad \Omega_s \leq |\Omega|$$

- Onde:
- Ω é o parâmetro de ondulação da banda de passagem
- Ω_p é a frequência de corte da banda passante em rad/seg
- A é o parâmetro de atenuação da banda de corte
- Ω_s é a frequência da banda de corte

ALGUMAS PRELIMINARES

- Especificações de um filtro passa-baixa analógico



$$|H_a(j\Omega_p)|^2 = \frac{1}{1+\epsilon^2} \quad \text{at } \Omega = \Omega_p$$

$$|H_a(j\Omega_s)|^2 = \frac{1}{A^2} \quad \text{at } \Omega = \Omega_s$$

ALGUMAS PRELIMINARES

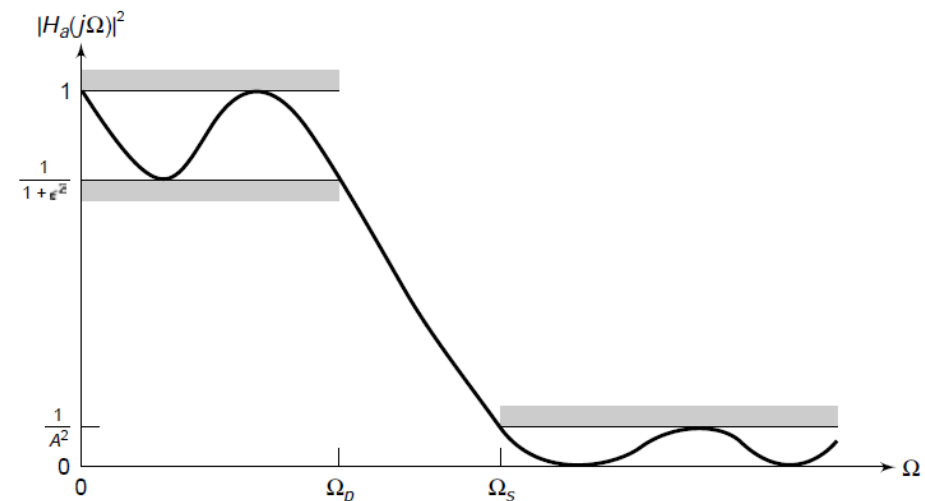
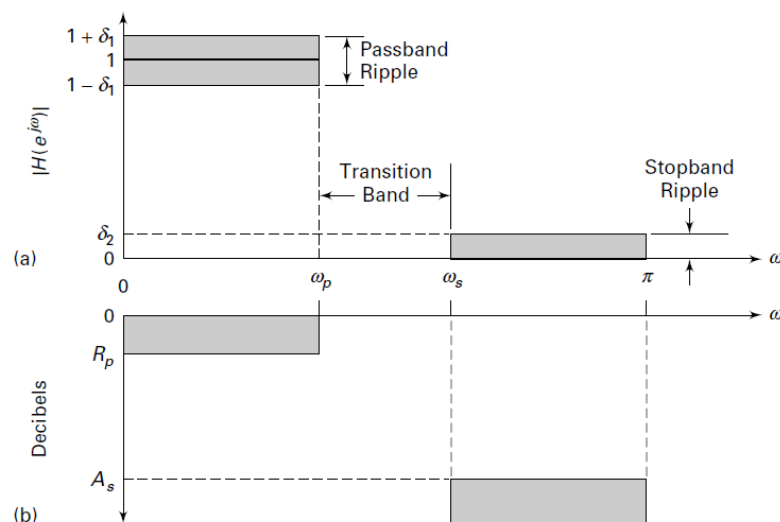
- Escala Relativa

- Os parâmetros $\hat{U}$ e A estão relacionados aos parâmetros R_p e A_s na escala dB como:

$$R_p = -10 \log_{10} \frac{1}{1 + \epsilon^2} \Rightarrow \epsilon = \sqrt{10^{R_p/10} - 1} \quad A_s = -10 \log_{10} \frac{1}{A^2} \Rightarrow A = 10^{A_s/20}$$

- As tolerâncias d_1 e d_2 da escala absoluta são relacionados a $\hat{U}$ e A por:

$$\frac{1 - \delta_1}{1 + \delta_1} = \sqrt{\frac{1}{1 + \epsilon^2}} \Rightarrow \epsilon = \frac{2\sqrt{\delta_1}}{1 - \delta_1} \quad \frac{\delta_2}{1 + \delta_1} = \frac{1}{A} \Rightarrow A = \frac{1 + \delta_1}{\delta_2}$$



ALGUMAS PRELIMINARES

- Propriedades de $|H_a(j\Omega)|^2$
 - Especificações de filtros analógicos não têm informação de fase
 - Para calcular a função de sistema $H_a(s)$ no domínio-s considere:

$$H_a(j\Omega) = H_a(s)|_{s=j\Omega}$$

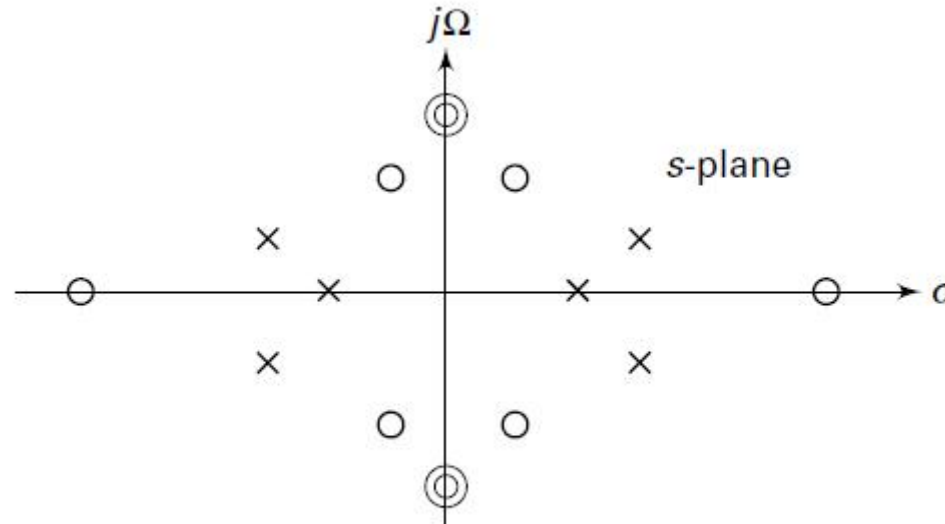
$$|H_a(j\Omega)|^2 = H_a(j\Omega)H_a^*(j\Omega) = H_a(j\Omega)H_a(-j\Omega) = H_a(s)H_a(-s)|_{s=j\Omega}$$

$$H_a(s)H_a(-s) = |H_a(j\Omega)|^2 \Big|_{\Omega=s/j}$$

- Portanto, os polos e zeros da função $|H_a(j\Omega)|^2$ são distribuídos em uma simetria imagem de espelho com respeito ao eixo $j\Omega$.
- Para filtros reais, polos e zeros ocorrem em pares complexos conjugados (ou simetria imagem de espelho com relação ao eixo real).
- $H_a(s)$ é a função de sistema de nosso filtro analógico.
- Queremos que $H_a(s)$ represente um filtro causal e estável. Então todos os polos de $H_a(s)$ devem estar dentro do semi-plano esquerdo. Assim designamos todos os polos no semi-planos esquerdo de $H_a(s)$ $H_a(-s)$ para $H_a(s)$.

ALGUMAS PRELIMINARES

- Entretanto, zeros de $H_a(s)$ podem existir em qualquer lugar no plano s . Portanto, eles não são unicamente determinados a menos que todos eles estejam no eixo $j\Omega$.
- Escolheremos os zeros de $H_a(s)H_a(-s)$ que estão no lado esquerdo ou no eixo $j\Omega$ como os zeros de $H_a(s)$. O filtro resultante é então chamado de filtro de fase mínima.



Typical pole-zero pattern of $H_a(s)H_a(-s)$

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- O projeto de filtros IIR reside na existência de filtros analógicos para obter filtros digitais.
- Esses filtros analógicos são chamados de filtros protótipos.
- Três protótipos são largamente usados na prática: Butterworth, Chebyshev (tipo I e II) e Elíptico.
- Vamos ver as características das versões passa-baixa desses filtros.

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

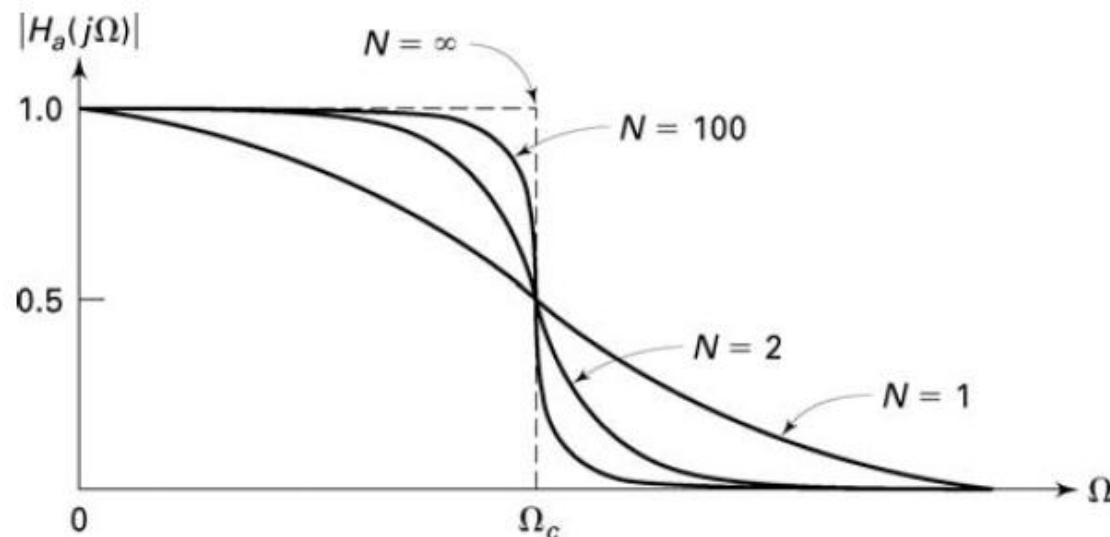
- Filtro Passa-Baixa Butterworth

- A principal característica desse filtro é que a resposta em magnitude é plana (flat) na banda de passagem e de corte.

- A resposta quadrática de magnitude de um FPB de N ésima ordem é dada por:

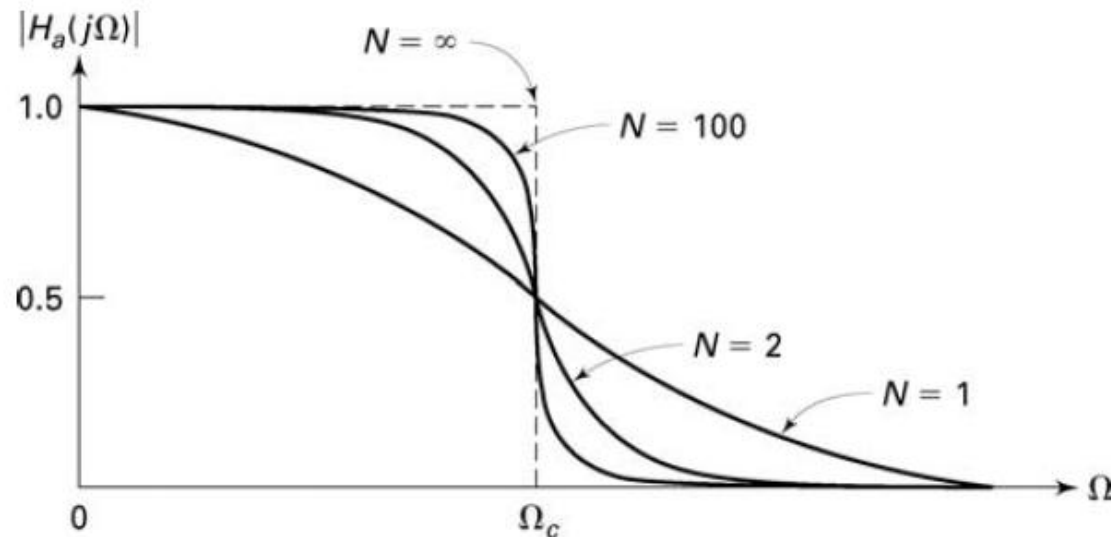
$$|H_a(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

- Onde N é a ordem do filtro e Ω_c é frequência de corte em rad/seg.



CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Filtro Passa-Baixa Butterworth



- Do gráfico, podemos observar:
 - i) Em $\Omega = 0$, $|H_a(j0)|^2 = 1$, para todo N .
 - ii) Em $\Omega = \Omega_c$, $|H_a(j\Omega_c)|^2 = 0,5$, para todo N , o que implica 3 dB de atenuação em Ω_c .
 - iii) $|H_a(j\Omega_c)|^2$ é uma função monotonicamente decrescente em Ω .
 - iv) $|H_a(j\Omega_c)|^2$ se aproxima de um FPB ideal em $N \rightarrow \infty$.
 - v) $|H_a(j\Omega_c)|^2$ é maximamente plano em $\Omega = 0$.

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Filtro Passa-Baixa Butterworth

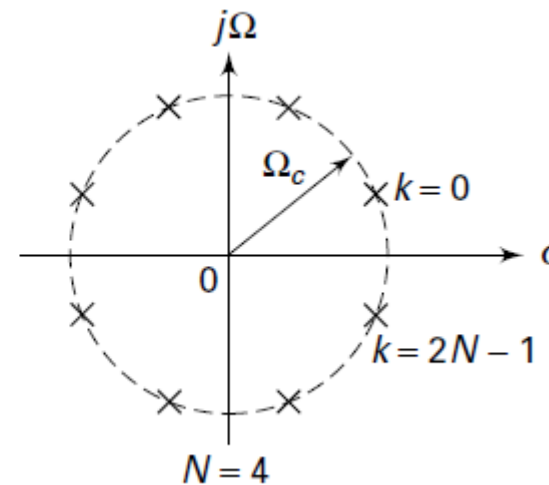
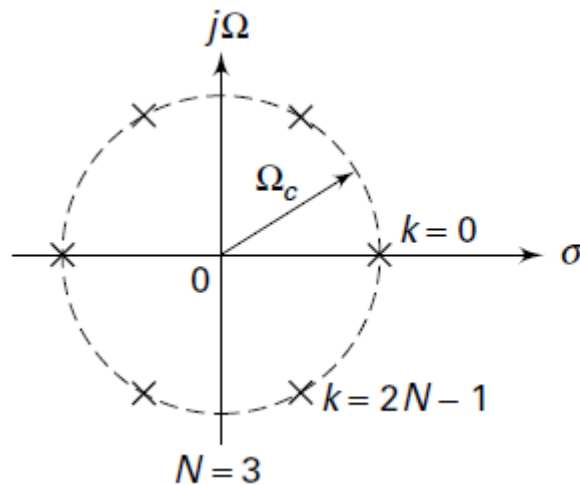
- Sua função de sistema $H_a(s)$ é:

$$H_a(s)H_a(-s) = |H_a(j\Omega)|^2 \Big|_{\Omega=s/j} = \frac{1}{1 + \left(\frac{s}{j\Omega_c}\right)^{2N}} = \frac{(j\Omega)^{2N}}{s^{2N} + (j\Omega_c)^{2N}}$$

- Para projetar o filtro, precisamos encontrar as raízes e pólos da função do sistema.
- Os pólos são dados por:
 - $p_k = (-1)^{1/2N} (j\Omega_c) = \Omega_c \cdot e^{j\pi(2k+N+1)/2N}$, $k = 0, 1, 2, \dots, 2N-1$
 - Assim, os pólos estão em um círculo de raio Ω_c com espaçamento angular de π/N radianos
 - Para N ímpar os polos são dados por $p_k = \Omega_c \cdot e^{j\pi k/N}$, $k = 0, 1, 2, \dots, 2N-1$
 - Para N par os polos são dados por $p_k = \Omega_c \cdot e^{j(\pi/2N + k\pi/N)}$, $k = 0, 1, 2, \dots, 2N-1$
 - Os polos são simetricamente localizados com respeito ao eixo $j\Omega$
 - Um polo nunca cai no eixo imaginário e cai no eixo real somente se N é ímpar

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Filtro Passa-Baixa Butterworth



- Um filtro causal e estável $H_a(s)$ pode ser especificado selecionando polos no semi-plano esquerdo e $H_a(s)$ pode ser escrito na forma

$$H_a(s) = \frac{\Omega_c^N}{\prod_{\text{LHP poles}} (s - p_k)}$$

□ **EXAMPLE 8.1** Given that $|H_a(j\Omega)|^2 = \frac{1}{1 + 64\Omega^6}$, determine the analog filter system function $H_a(s)$.

Solution From the given magnitude-squared response,

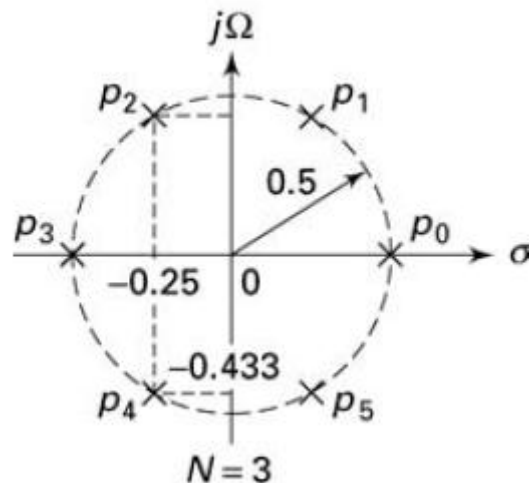
$$|H_a(j\Omega)|^2 = \frac{1}{1 + 64\Omega^6} = \frac{1}{1 + \left(\frac{\Omega}{0.5}\right)^{2(3)}}$$

Comparing this with expression (8.45), we obtain $N = 3$ and $\Omega_c = 0.5$. The poles of $H_a(s)H_a(-s)$ are as shown in Figure 8.14.

Hence

$$\begin{aligned} H_a(j\Omega) &= \frac{\Omega_c^3}{(s - p_2)(s - p_3)(s - p_4)} \\ &= \frac{1/8}{(s + 0.25 - j0.433)(s + 0.5)(s + 0.25 + j0.433)} \\ &= \frac{0.125}{(s + 0.5)(s^2 + 0.5s + 0.25)} \end{aligned}$$

□



```

function [b,a] = u_buttap(N,Omegac);
% Unnormalized Butterworth Analog Lowpass Filter Prototype
% -----
% [b,a] = u_buttap(N,Omegac);
%      b = numerator polynomial coefficients of Ha(s)
%      a = denominator polynomial coefficients of Ha(s)
%      N = Order of the Butterworth Filter
%      Omegac = Cutoff frequency in radians/sec
%
[z,p,k] = buttap(N);
      p = p*Omegac;
      k = k*Omegac^N;
      B = real(poly(z));
      b0 = k;  b = k*B;  a = real(poly(p));

```



```

function [C,B,A] = sdir2cas(b,a);
% DIRECT-form to CASCADE-form conversion in s-plane
% -----
% [C,B,A] = sdir2cas(b,a)
% C = gain coefficient
% B = K by 3 matrix of real coefficients containing bk's
% A = K by 3 matrix of real coefficients containing ak's
% b = numerator polynomial coefficients of DIRECT form
% a = denominator polynomial coefficients of DIRECT form
%
Na = length(a)-1; Nb = length(b)-1;
% compute gain coefficient C
b0 = b(1); b = b/b0; a0 = a(1); a = a/a0; C = b0/a0;
%
% Denominator second-order sections:
p = cplxpair(roots(a)); K = floor(Na/2);
if K*2 == Na % Computation when Na is even
    A = zeros(K,3);
    for n=1:2:Na
        Arow = p(n:1:n+1,:); Arow = poly(Arow);
        A(fix((n+1)/2),:) = real(Arow);
    end
elseif Na == 1 % Computation when Na = 1
    A = [0 real(poly(p))];
else % Computation when Na is odd and > 1
    A = zeros(K+1,3);
    for n=1:2:2*K
        Arow = p(n:1:n+1,:); Arow = poly(Arow);
        A(fix((n+1)/2),:) = real(Arow);
    end
    A(K+1,:) = [0 real(poly(p(Na)))];
end

% Numerator second-order sections:
z = cplxpair(roots(b)); K = floor(Nb/2);
if Nb == 0 % Computation when Nb = 0
    B = [0 0 poly(z)];
elseif K*2 == Nb % Computation when Nb is even
    B = zeros(K,3);
    for n=1:2:Nb
        Brow = z(n:1:n+1,:); Brow = poly(Brow);
        B(fix((n+1)/2),:) = real(Brow);
    end
elseif Nb == 1 % Computation when Nb = 1
    B = [0 real(poly(z))];
else % Computation when Nb is odd and > 1
    B = zeros(K+1,3);
    for n=1:2:2*K
        Brow = z(n:1:n+1,:); Brow = poly(Brow);
        B(fix((n+1)/2),:) = real(Brow);
    end
    B(K+1,:) = [0 real(poly(z(Nb)))];
end

```

□ **EXAMPLE 8.2** Design a 3rd-order Butterworth analog prototype filter with $\Omega_c = 0.5$ given in Example 8.1.

Solution

MATLAB script:

```

>> N = 3; OmegaC = 0.5; [b,a] = u_buttap(N,OmegaC);
>> [C,B,A] = sdir2cas(b,a)
C = 0.1250
B = 0      0      1
A = 1.0000  0.5000  0.2500
      0      1.0000  0.5000

```

The cascade form coefficients agree with those in Example 8.1. □

$$H_a(j\Omega) = \frac{0.125}{(s + 0.5)(s^2 + 0.5s + 0.25)}$$

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Filtro Passa-Baixa Butterworth
- O FPB analógico é especificado pelos parâmetros Ω_p , Ω_s , R_p e A_s
- Assim, a essência do projeto no caso do filtro de Butterworth é obter a ordem N e a frequência de corte dada Ω_c .

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- at $\Omega = \Omega_p$, $-10 \log_{10} |H_a(j\Omega)|^2 = R_p$ or

$$-10 \log_{10} \left(\frac{1}{1 + \left(\frac{\Omega_p}{\Omega_c} \right)^{2N}} \right) = R_p$$

and

- at $\Omega = \Omega_s$, $-10 \log_{10} |H_a(j\Omega)|^2 = A_s$ or

$$-10 \log_{10} \left(\frac{1}{1 + \left(\frac{\Omega_s}{\Omega_c} \right)^{2N}} \right) = A_s$$

Solving these two equations for N and Ω_c , we have

$$N = \frac{\log_{10} [(10^{R_p/10} - 1) / (10^{A_s/10} - 1)]}{2 \log_{10} (\Omega_p / \Omega_s)}$$

In general, N will not be an integer. Since we want N to be an integer, we must choose

$$N = \left\lceil \frac{\log_{10} [(10^{R_p/10} - 1) / (10^{A_s/10} - 1)]}{2 \log_{10} (\Omega_p / \Omega_s)} \right\rceil \quad (8.49)$$

Escolha o menor inteiro maior que. Ex $[4.5]=5$

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Obviamente, isso irá gerar um filtro com ordem maior do que o necessário
- Para satisfazer exatamente as especificações do projeto em Ω_p

$$\Omega_c = \frac{\Omega_p}{2N \sqrt{(10^{R_p/10} - 1)}}$$

- Para satisfazer exatamente as especificações do projeto em Ω_s

$$\Omega_c = \frac{\Omega_s}{2N \sqrt{(10^{A_s/10} - 1)}}$$

□ **EXAMPLE 8.3** Design a lowpass Butterworth filter to satisfy

Passband cutoff: $\Omega_p = 0.2\pi$; Passband ripple: $R_p = 7\text{dB}$

Stopband cutoff: $\Omega_s = 0.3\pi$; Stopband ripple: $A_s = 16\text{dB}$

Solution

From (8.49)

$$N = \left\lceil \frac{\log_{10} [(10^{0.7} - 1) / (10^{1.6} - 1)]}{2 \log_{10} (0.2\pi / 0.3\pi)} \right\rceil = \lceil 2.79 \rceil = 3$$

To satisfy the specifications exactly at Ω_p , from (8.50) we obtain

$$\Omega_c = \frac{0.2\pi}{\sqrt[6]{(10^{0.7} - 1)}} = 0.4985$$

To satisfy specifications exactly at Ω_s , from (8.51) we obtain

$$\Omega_c = \frac{0.3\pi}{\sqrt[6]{(10^{1.6} - 1)}} = 0.5122$$

Now we can choose any Ω_c between the above two numbers. Let us choose $\Omega_c = 0.5$. We have to design a Butterworth filter with $N = 3$ and $\Omega_c = 0.5$, which we did in Example 8.1. Hence

$$H_a(j\Omega) = \frac{0.125}{(s + 0.5)(s^2 + 0.5s + 0.25)}$$

□



Técnicas de Projeto de Filtros IIR

Características de Protótipos Analógicos

- Filtro de Butterworth: Exemplo

- Solução:

- Podemos escolher Ω_c entre esses dois valores, por exemplo $\Omega_c = 0,5$
 - Temos que projetar um filtro Butterworth com $N = 3$ e $\Omega_c = 0,5$

- Ou seja: $|H_a(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{0,5}\right)^6}$

- Como $\Omega = s/j$, temos:

$$|H_a(s)|^2 = \frac{1}{1 + \left(\frac{s}{0,5j}\right)^6} = \frac{1}{1 - 2^6 s^6} = \frac{1}{1 - 64s^6}$$

Técnicas de Projeto de Filtros IIR

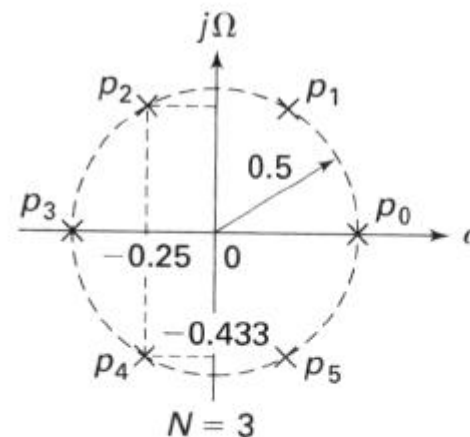
Características de Protótipos Analógicos

■ Filtro de Butterworth: Exemplo

■ Solução:

- Os pólos p_k da função anterior podem ser calculados no MatLab como:

- `>> a = [-64 0 0 0 0 0 1];`
- `>> b = roots(a)`
- `b =`
- -0.5000
- -0.2500 + 0.4330i
- -0.2500 - 0.4330i
- 0.5000
- 0.2500 + 0.4330i
- 0.2500 - 0.4330i





Técnicas de Projeto de Filtros IIR

Características de Protótipos Analógicos

- Filtro de Butterworth: Exemplo
 - Solução:
 - Para termos um filtro causal e estável, usamos os pólos do semi-plano esquerdo:

$$H_a(s) = \frac{1}{(s + 0,5)(s + 0,25 + j0,433)(s + 0,25 - j0,433)}$$

$$H_a(s) = \frac{1}{(s + 0,5)(s^2 + 0,5s + 0,25)}$$



Técnicas de Projeto de Filtros IIR

Características de Protótipos Analógicos

- Filtro de Butterworth: Exemplo

- Solução:

- Vamos ajustar o numerador para que o ganho na frequência zero seja unitário

- Ou seja, no denominador, quando $s = 0$, temos:

- $(s + 0,5)(s^2 + 0,5s + 0,25) = 0,5 \cdot 0,25 = 0,125$

- Logo, o denominador é multiplicado por um fator de $1/8$ e temos:

$$H_a(s) = \frac{1/8}{(s + 0,5)(s^2 + 0,5s + 0,25)} = \frac{0,125}{(s + 0,5)(s^2 + 0,5s + 0,25)}$$


```

function [b,a] = afd_butt(Wp,Ws,Rp,As);
% Analog Lowpass Filter Design: Butterworth
% -----
% [b,a] = afd_butt(Wp,Ws,Rp,As);
% b = Numerator coefficients of Ha(s)
% a = Denominator coefficients of Ha(s)
% Wp = Passband edge frequency in rad/sec; Wp > 0
% Ws = Stopband edge frequency in rad/sec; Ws > Wp > 0
% Rp = Passband ripple in +dB; (Rp > 0)
% As = Stopband attenuation in +dB; (As > 0)
%
if Wp <= 0
    error('Passband edge must be larger than 0')
end
if Ws <= Wp
    error('Stopband edge must be larger than Passband edge')
end
if (Rp <= 0) | (As < 0)
    error('PB ripple and/or SB attenuation ust be larger than 0')
end

N = ceil((log10((10^(Rp/10)-1)/(10^(As/10)-1)))/(2*log10(Wp/Ws)));
fprintf('\n*** Butterworth Filter Order = %2.0f \n',N)
OmegaC = Wp/((10^(Rp/10)-1)^(1/(2*N)));
[b,a]=u_buttap(N,OmegaC);

```

```

function [db,mag,pha,w] = freqs_m(b,a,wmax);
% Computation of s-domain frequency response: Modified version
% -----
% [db,mag,pha,w] = freqs_m(b,a,wmax);

% db = Relative magnitude in db over [0 to wmax]
% mag = Absolute magnitude over [0 to wmax]
% pha = Phase response in radians over [0 to wmax]
% w = array of 500 frequency samples between [0 to wmax]
% b = Numerator polynomial coefficients of Ha(s)
% a = Denominator polynomial coefficients of Ha(s)
% wmax = Maximum frequency in rad/sec over which response is desired
%
w = [0:1:500]*wmax/500; H = freqs(b,a,w);
mag = abs(H); db = 20*log10((mag+eps)/max(mag)); pha = angle(H);

```

- **EXAMPLE 8.4** Design the analog Butterworth lowpass filter specified in Example 8.3 using MATLAB.

Solution

MATLAB script:

```
>> Wp = 0.2*pi; Ws = 0.3*pi; Rp = 7; As = 16;
>> Ripple = 10 ^ (-Rp/20); Attn = 10 ^ (-As/20);
>> % Analog filter design:
>> [b,a] = afd_butt(Wp,Ws,Rp,As);
*** Butterworth Filter Order = 3
>> % Calculation of second-order sections:
>> [C,B,A] = sdir2cas(b,a)
C = 0.1238
B = 0      0      1
A = 1.0000  0.4985  0.2485
      0      1.0000  0.4985
>> % Calculation of Frequency Response:
>> [db,mag,pha,w] = freqs_m(b,a,0.5*pi);
>> % Calculation of Impulse response:
>> [ha,x,t] = impulse(b,a);
```

The system function is given by

$$H_a(s) = \frac{0.1238}{(s^2 + 0.4985s + 0.2485)(s + 0.4985)}$$

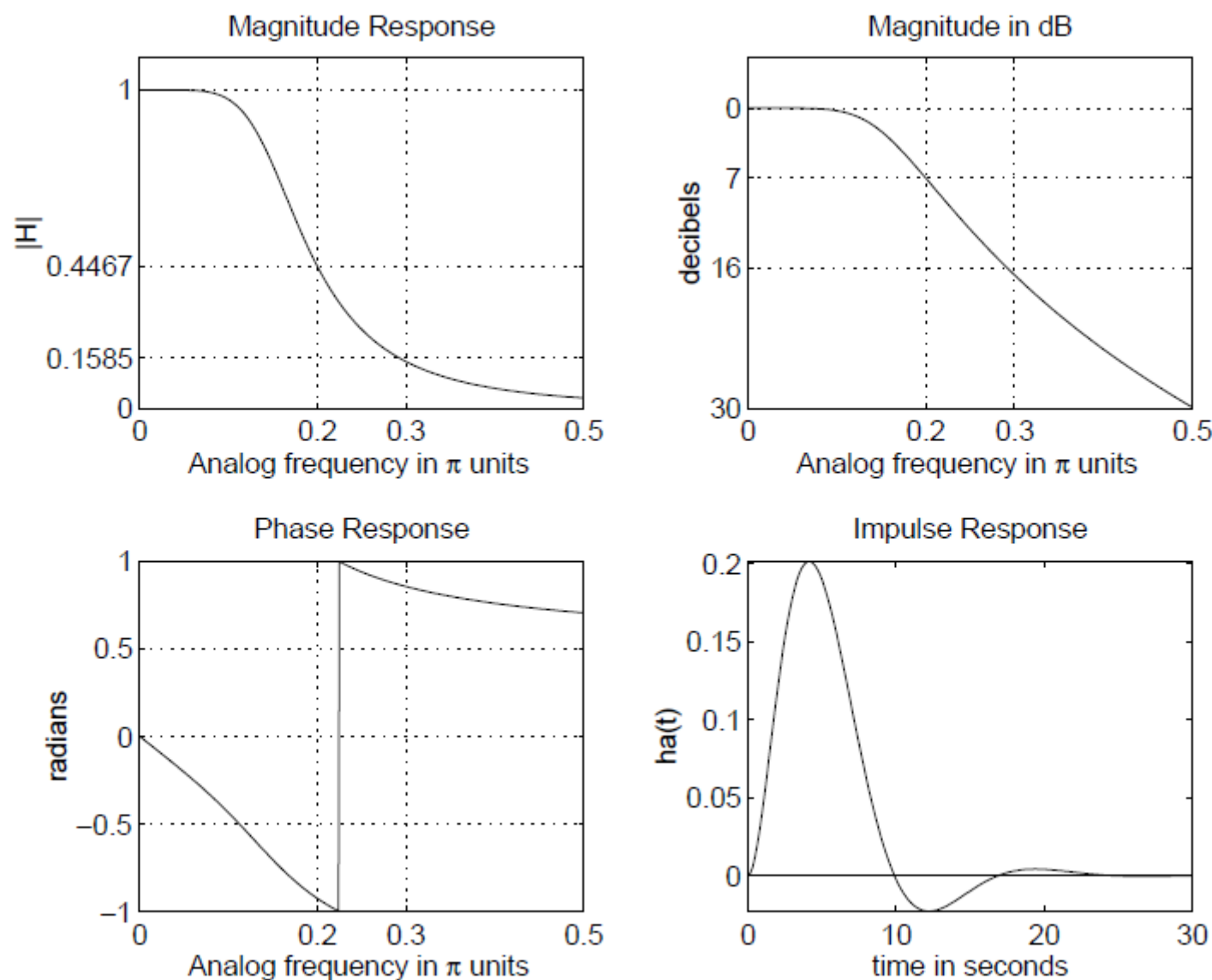


FIGURE 8.15 *Butterworth analog filter in Example 8.4*



Técnicas de Projeto de Filtros IIR

Características de Protótipos Analógicos

- Filtro de Butterworth: Exemplo
 - Solução:
 - Para transformar o filtro em digital, podemos usar o método de transformação bilinear
 - Nele, consideramos:

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}} \Rightarrow z = \frac{1 + sT/2}{1 - sT/2}$$

- onde T é um parâmetro

OBS: Explicação nas notas de aula.



Técnicas de Projeto de Filtros IIR

Características de Protótipos Analógicos

- Filtro de Butterworth: Exemplo

- Solução:

- No nosso caso, consideramos $T = 1$:

$$s = 2 \frac{1 - z^{-1}}{1 + z^{-1}} \Rightarrow H_a(z) = H_a(s) \Big|_{s=2 \frac{1 - z^{-1}}{1 + z^{-1}}} =$$

$$H_a(z) = \frac{0,125}{\left[2 \frac{1 - z^{-1}}{1 + z^{-1}} + 0,5 \right] \left[\left(2 \frac{1 - z^{-1}}{1 + z^{-1}} \right)^2 + 0,5 \left(2 \frac{1 - z^{-1}}{1 + z^{-1}} \right) + 0,25 \right]}$$

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Filtro Chebyshev
 - Existem dois tipos: Chebyshev tipo I e Chebyshev tipo II
 - Chebyshev tipo I tem equiripple na banda passante
 - Chebyshev tipo II tem equiripple na banda de corte
 - Um filtro Chebyshev tem menor ordem que um Butterworth para as mesmas especificações



Técnicas de Projeto de Filtros IIR

Características de Protótipos Analógicos

- Filtro de Chebyshev

- A resposta quadrática de magnitude de um filtro Chebyshev tipo I é dada por:

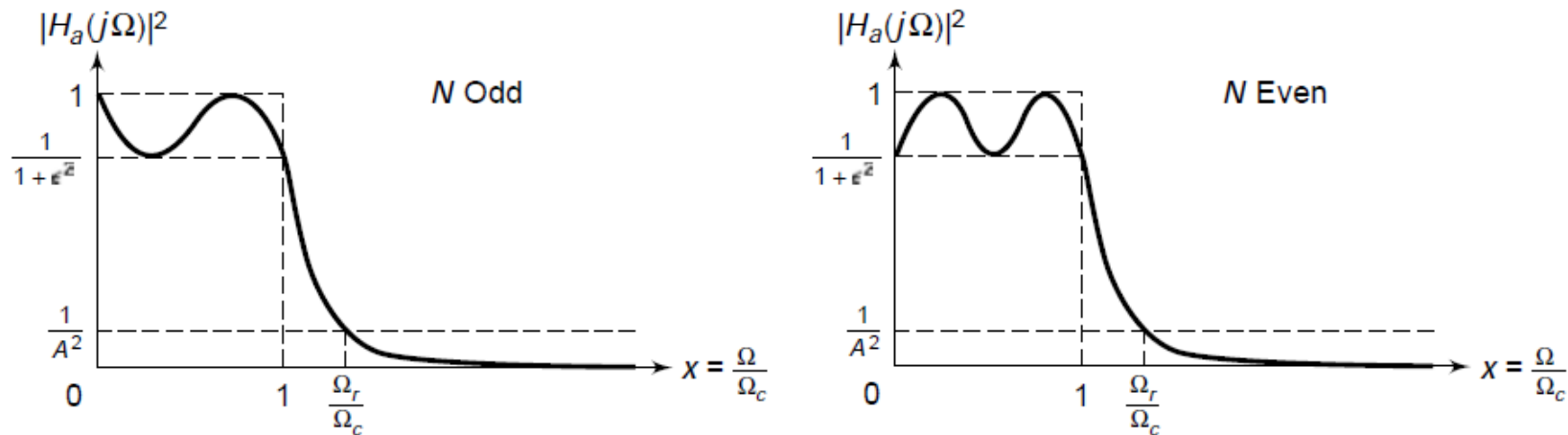
$$|H_a(j\Omega)|^2 = \frac{1}{1 + \varepsilon^2 T_N^2\left(\frac{\Omega}{\Omega_c}\right)}$$

- onde N é a ordem do filtro, ε é o fator de ondulação da banda de passagem e $T_N(x)$ é o polinômio de Chebyshev dado por (podemos considerar $x = \Omega/\Omega_c$):

$$T_N(x) = \begin{cases} \cos(N \cdot \cos^{-1}(x)) & , \quad 0 \leq x \leq 1 \\ \cosh(\cosh^{-1}(x)) & , \quad 1 < x < \infty \end{cases}$$

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

• Filtro Chebyshev tipo I



- At $x = 0$ (or $\Omega = 0$); $|H_a(j0)|^2 = 1$ for N odd.
 $|H_a(j0)|^2 = \frac{1}{1+\epsilon^2}$ for N even.
- At $x = 1$ (or $\Omega = \Omega_c$); $|H_a(j1)|^2 = \frac{1}{1+\epsilon^2}$ for all N .
- For $0 \leq x \leq 1$ (or $0 \leq \Omega \leq \Omega_c$), $|H_a(jx)|^2$ oscillates between 1 and $\frac{1}{1+\epsilon^2}$.
- For $x > 1$ (or $\Omega > \Omega_c$), $|H_a(jx)|^2$ decreases monotonically to 0.
- At $x = \Omega_r$, $|H_a(jx)|^2 = \frac{1}{A^2}$.

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Filtro Chebyshev tipo I

- Os polos de $H_a(s)H_a(-s)$ são obtidos encontrando as raízes de:

$$1 + \epsilon^2 T_N^2 \left(\frac{s}{j\Omega_c} \right)$$

- A solução desta equação é tediosa e difícil de se obter. Pode ser mostrado que se $p_k = \sigma_k + j\Omega_k$, $k = 0, \dots, N-1$ são as raízes (semi-plano esquerdo) do polinômio acima, então:

$$\sigma_k = (a\Omega_c) \cos \left[\frac{\pi}{2} + \frac{(2k+1)\pi}{2N} \right] \quad k = 0, \dots, N-1$$

$$\Omega_k = (b\Omega_c) \sin \left[\frac{\pi}{2} + \frac{(2k+1)\pi}{2N} \right]$$

$$a = \frac{1}{2} \left(\sqrt[N]{\alpha} - \sqrt[N]{1/\alpha} \right), \quad b = \frac{1}{2} \left(\sqrt[N]{\alpha} + \sqrt[N]{1/\alpha} \right), \quad \text{and} \quad \alpha = \frac{1}{\epsilon} + \sqrt{1 + \frac{1}{\epsilon^2}}$$

- A função de sistema é dada por:

$$H_a(s) = \frac{K}{\prod_k (s - p_k)}$$

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Filtro Chebyshev tipo I
 - A função de sistema é dada por:

$$H_a(s) = \frac{K}{\prod_k (s - p_k)}$$

- K é um fator normalizante escolhido para fazer:

$$H_a(j0) = \begin{cases} 1, & N \text{ odd} \\ \frac{1}{\sqrt{1 + \epsilon^2}}, & N \text{ even} \end{cases}$$

```

function [b,a] = u_chb1ap(N,Rp,Omegac);
% Unnormalized Chebyshev-1 Analog Lowpass Filter Prototype
% -----
% [b,a] = u_chb1ap(N,Rp,Omegac);
%      b = numerator polynomial coefficients
%      a = denominator polynomial coefficients
%      N = Order of the Elliptic Filter
%      Rp = Passband Ripple in dB; Rp > 0
%      Omegac = Cutoff frequency in radians/sec
%
[z,p,k] = cheb1ap(N,Rp);  a = real(poly(p));  aNn = a(N+1);
      p = p*Omegac;  a = real(poly(p));  aNu = a(N+1);
      k = k*aNu/aNn;
      b0 = k;  B = real(poly(z));  b = k*B;

```

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Filtro Chebyshev tipo I
 - Dado Ω_p , Ω_s , R_p e A_s , três parâmetros são exigidos para determinar um filtro Chebyshev I: ϵ , Ω_c e N .

$$\epsilon = \sqrt{10^{0.1R_p} - 1} \quad \text{and} \quad A = 10^{A_s/20} \quad \Omega_c = \Omega_p \quad \text{and} \quad \Omega_r = \frac{\Omega_s}{\Omega_p}$$

$$g = \sqrt{(A^2 - 1)/\epsilon^2} \quad N = \left\lceil \frac{\log_{10} [g + \sqrt{g^2 - 1}]}{\log_{10} [\Omega_r + \sqrt{\Omega_r^2 - 1}]} \right\rceil$$

□ **EXAMPLE 8.5** Design a lowpass Chebyshev-I filter to satisfy

Passband cutoff: $\Omega_p = 0.2\pi$; Passband ripple: $R_p = 1\text{dB}$

Stopband cutoff: $\Omega_s = 0.3\pi$; Stopband ripple: $A_s = 16\text{dB}$

Solution

First compute the necessary parameters.

$$\epsilon = \sqrt{10^{0.1(1)} - 1} = 0.5088 \quad A = 10^{16/20} = 6.3096$$

$$\Omega_c = \Omega_p = 0.2\pi \quad \Omega_r = \frac{0.3\pi}{0.2\pi} = 1.5$$

$$g = \sqrt{(A^2 - 1)/\epsilon^2} = 12.2429 \quad N = 4$$

Now we can determine $H_a(s)$.

$$\alpha = \frac{1}{\epsilon} + \sqrt{1 + \frac{1}{\epsilon^2}} = 4.1702$$

$$a = 0.5 \left(\sqrt[N]{\alpha} - \sqrt[N]{1/\alpha} \right) = 0.3646$$

$$b = 0.5 \left(\sqrt[N]{\alpha} + \sqrt[N]{1/\alpha} \right) = 1.0644$$

There are four poles for $H_a(s)$:

$$p_{0,3} = (a\Omega_c) \cos \left[\frac{\pi}{2} + \frac{\pi}{8} \right] \pm (b\Omega_c) \sin \left[\frac{\pi}{2} + \frac{\pi}{8} \right] = -0.0877 \pm j0.6179$$

$$p_{1,2} = (a\Omega_c) \cos \left[\frac{\pi}{2} + \frac{3\pi}{8} \right] \pm (b\Omega_c) \sin \left[\frac{\pi}{2} + \frac{3\pi}{8} \right] = -0.2117 \pm j0.2559$$

Hence

$$H_a(s) = \frac{K}{\prod_{k=0}^3 (s - p_k)} = \frac{\overbrace{0.89125 \times .1103 \times .3895}^{0.03829}}{(s^2 + 0.1754s + 0.3895)(s^2 + 0.4234s + 0.1103)}$$

Note that the numerator is such that

$$H_a(j0) = \frac{1}{\sqrt{1 + \epsilon^2}} = 0.89125$$

□


```

function [b,a] = afd_chb1(Wp,Ws,Rp,As);
% Analog Lowpass Filter Design: Chebyshev-1
% -----
% [b,a] = afd_chb1(Wp,Ws,Rp,As);
% b = Numerator coefficients of Ha(s)
% a = Denominator coefficients of Ha(s)
% Wp = Passband edge frequency in rad/sec; Wp > 0
% Ws = Stopband edge frequency in rad/sec; Ws > Wp > 0
% Rp = Passband ripple in +dB; (Rp > 0)
% As = Stopband attenuation in +dB; (As > 0)
%
if Wp <= 0
    error('Passband edge must be larger than 0')
end
if Ws <= Wp
    error('Stopband edge must be larger than Passband edge')
end
if (Rp <= 0) | (As < 0)
    error('PB ripple and/or SB attenuation ust be larger than 0')
end

ep = sqrt(10^(Rp/10)-1); A = 10^(As/20);
OmegaC = Wp; OmegaR = Ws/Wp; g = sqrt(A*A-1)/ep;
N = ceil(log10(g+sqrt(g*g-1))/log10(OmegaR+sqrt(OmegaR*OmegaR-1)));
fprintf('\n*** Chebyshev-1 Filter Order = %2.0f \n',N)
[b,a]=u_chblap(N,Rp,OmegaC);

```


□ **EXAMPLE 8.5** Design a lowpass Chebyshev-I filter to satisfy

Passband cutoff: $\Omega_p = 0.2\pi$; Passband ripple: $R_p = 1\text{dB}$

Stopband cutoff: $\Omega_s = 0.3\pi$; Stopband ripple: $A_s = 16\text{dB}$

□ **EXAMPLE 8.6** Design the analog Chebyshev-I lowpass filter given in Example 8.5 using MATLAB.

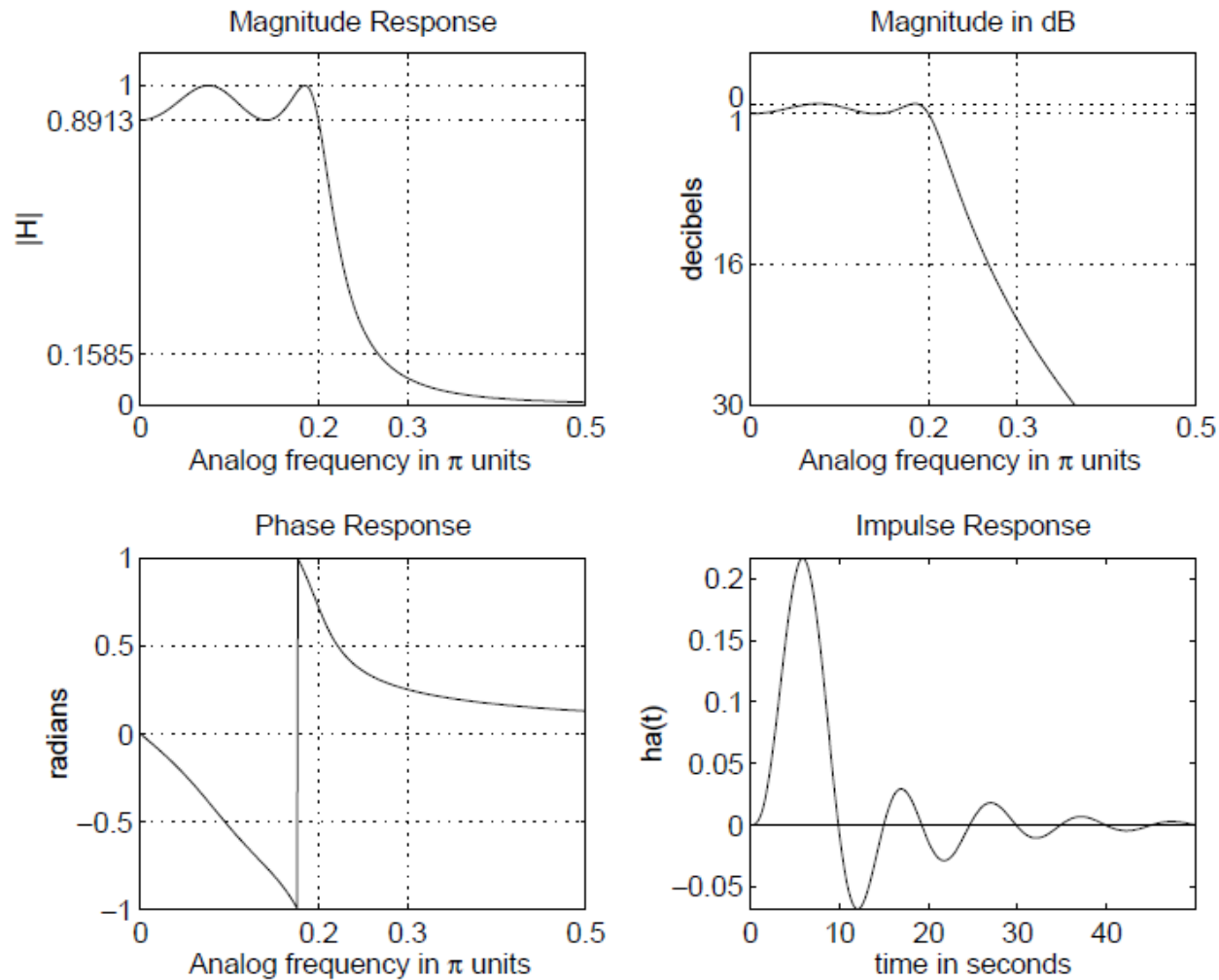
Solution

MATLAB script:

```
>> Wp = 0.2*pi; Ws = 0.3*pi; Rp = 1; As = 16;
>> Ripple = 10 ^ (-Rp/20); Attn = 10 ^ (-As/20);
>> % Analog filter design:
>> [b,a] = afd_chb1(Wp,Ws,Rp,As);
*** Chebyshev-1 Filter Order = 4
>> % Calculation of second-order sections:
>> [C,B,A] = sdir2cas(b,a)
C = 0.0383
B = 0      0      1
A = 1.0000    0.4233    0.1103
    1.0000    0.1753    0.3895
>> % Calculation of Frequency Response:
>> [db,mag,pha,w] = freqs_m(b,a,0.5*pi);
>> % Calculation of Impulse response:
>> [ha,x,t] = impulse(b,a);
```

The specifications are satisfied by a 4th-order Chebyshev-I filter whose system function is

$$H_a(s) = \frac{0.0383}{(s^2 + 4233s + 0.1103)(s^2 + 0.1753s + 0.3895)}$$



CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Filtro Chebyshev tipo II

$$|H_a(j\Omega)|^2 = \frac{1}{1 + [\epsilon^2 T_N^2(\Omega_c/\Omega)]^{-1}}$$

```
function [b,a] = u_chb2ap(N,As,Omegac);  
% Unnormalized Chebyshev-2 Analog Lowpass Filter Prototype  
% -----  
% [b,a] = u_chb2ap(N,As,Omegac);  
%     b = numerator polynomial coefficients  
%     a = denominator polynomial coefficients  
%     N = Order of the Elliptic Filter  
%     As = Stopband Ripple in dB; As > 0  
% Omegac = Cutoff frequency in radians/sec  
%  
[z,p,k] = cheb2ap(N,As);  
    a = real(poly(p));  aNn = a(N+1);  
    p = p*Omegac;  a = real(poly(p));  aNu = a(N+1);  
    b = real(poly(z));  M = length(b);  bNn = b(M);  
    z = z*Omegac;  b = real(poly(z));  bNu = b(M);  
    k = k*(aNu*bNn)/(aNn*bNu);  
    b0 = k;  b = k*b;
```

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Filtro Chebyshev tipo II

```
function [b,a] = afd_chb2(Wp,Ws,Rp,As);  
% Analog Lowpass Filter Design: Chebyshev-2  
% -----  
% [b,a] = afd_chb2(Wp,Ws,Rp,As);  
% b = Numerator coefficients of Ha(s)  
% a = Denominator coefficients of Ha(s)  
% Wp = Passband edge frequency in rad/sec; Wp > 0  
% Ws = Stopband edge frequency in rad/sec; Ws > Wp > 0  
% Rp = Passband ripple in +dB; (Rp > 0)  
% As = Stopband attenuation in +dB; (As > 0)  
%  
if Wp <= 0  
    error('Passband edge must be larger than 0')  
end  
if Ws <= Wp  
    error('Stopband edge must be larger than Passband edge')  
end  
if (Rp <= 0) | (As < 0)  
    error('PB ripple and/or SB attenuation must be larger than 0')  
end  
  
ep = sqrt(10^(Rp/10)-1); A = 10^(As/20);  
OmegaC = Wp; OmegaR = Ws/Wp; g = sqrt(A*A-1)/ep;  
N = ceil(log10(g+sqrt(g*g-1))/log10(OmegaR+sqrt(OmegaR*OmegaR-1)));  
fprintf('\n*** Chebyshev-2 Filter Order = %2.0f \n',N)  
[b,a]=u_chb2ap(N,As,Ws);
```


□ **EXAMPLE 8.7** Design a Chebyshev-II analog lowpass filter to satisfy the specifications given in Example 8.5:

Passband cutoff: $\Omega_p = 0.2\pi$; Passband ripple: $R_p = 1\text{dB}$

Stopband cutoff: $\Omega_s = 0.3\pi$; Stopband ripple: $A_s = 16\text{dB}$

Solution

MATLAB script:

```
>> Wp = 0.2*pi; Ws = 0.3*pi; Rp = 1; As = 16;
>> Ripple = 10 ^ (-Rp/20); Attn = 10 ^ (-As/20);
>> % Analog filter design:
>> [b,a] = afd_chb2(Wp,Ws,Rp,As);
*** Chebyshev-2 Filter Order = 4
>> % Calculation of second-order sections:
>> [C,B,A] = sdir2cas(b,a)
C = 0.1585
B = 1.0000      0      6.0654
    1.0000      0      1.0407
A = 1.0000    1.9521    1.4747
    1.0000    0.3719    0.6784
>> % Calculation of Frequency Response:
>> [db,mag,pha,w] = freqs_m(b,a,0.5*pi);
>> % Calculation of Impulse response:
>> [ha,x,t] = impulse(b,a);
```

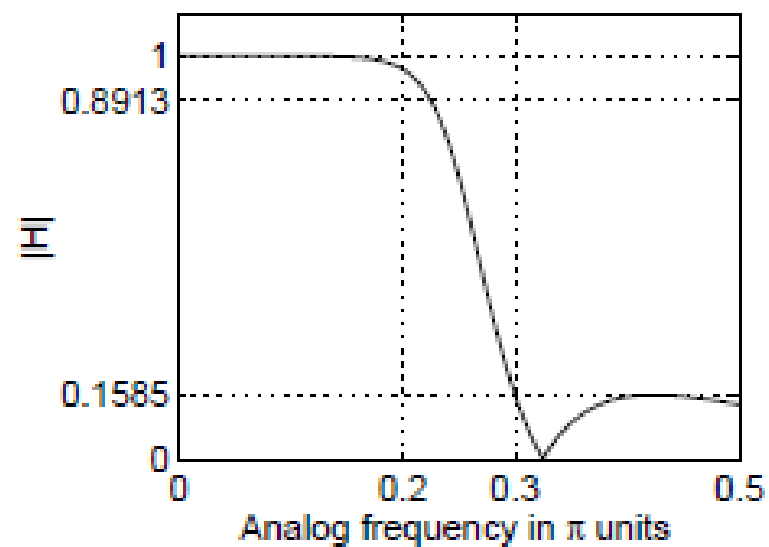
The specifications are satisfied by a 4th-order Chebyshev-II filter whose system function is

$$H_a(s) = \frac{0.1585 (s^2 + 6.0654) (s^2 + 1.0407)}{(s^2 + 1.9521s + 1.4747) (s^2 + 0.3719s + 0.6784)}$$

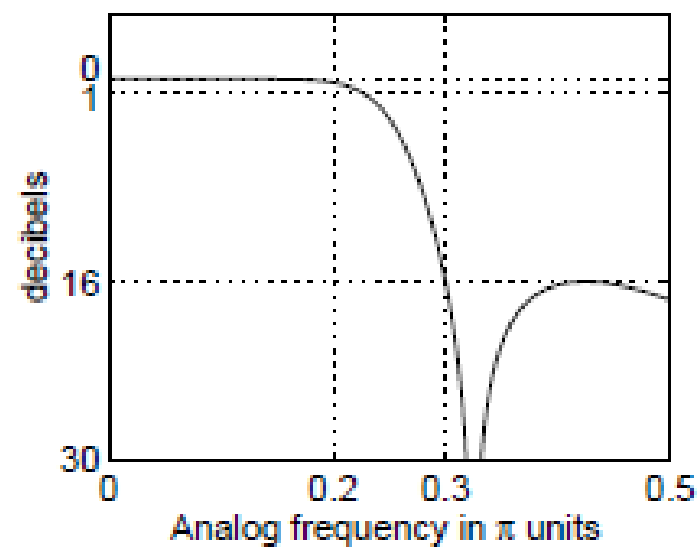
The filter plots are shown in Figure 8.17.

□

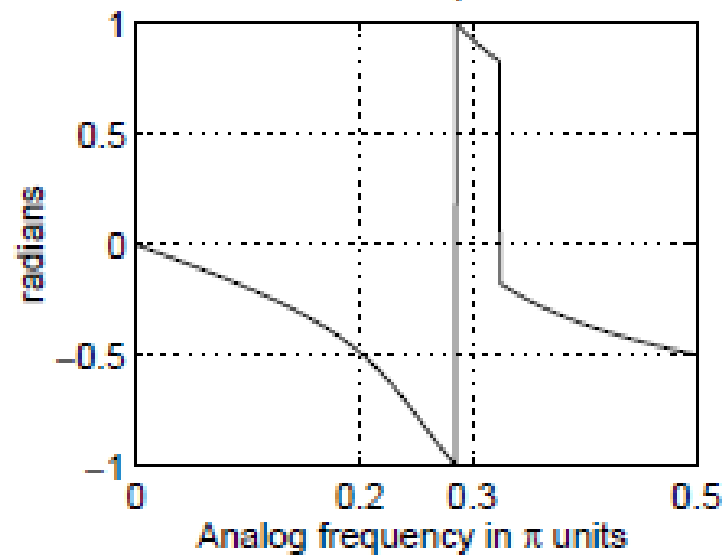
Magnitude Response



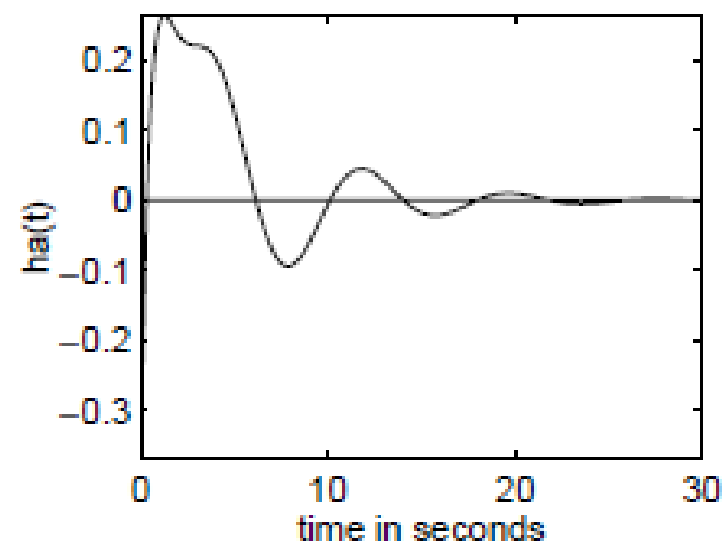
Magnitude in dB



Phase Response



Impulse Response



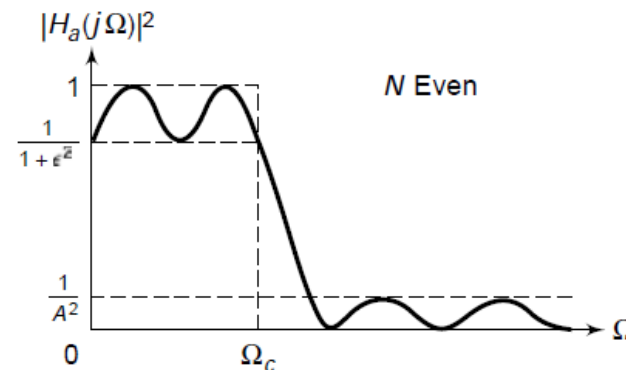
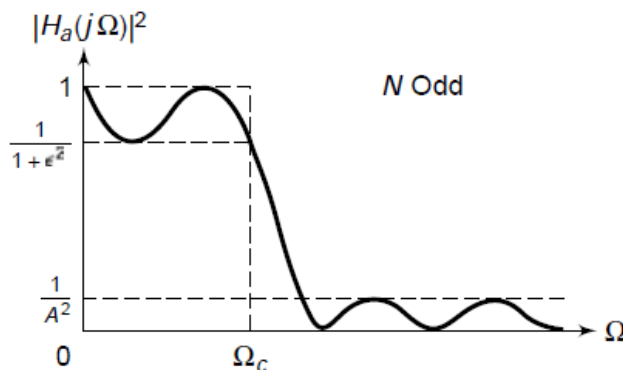
CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Filtro Elíptico

- Possuem equiripple na banda passante e de corte
- Alcançam a mínima ordem N para uma dada especificação, ou seja, alcançam a mais estreita banda de transição para uma dada ordem N.
- São difíceis de analisar e implementar.
- A resposta em magnitude é dada por:

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \epsilon^2 U_N^2 \left(\frac{\Omega}{\Omega_c} \right)}$$

- Onde U_N é a função Jacobiana Elíptica de ordem N ésima



CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Filtro Elíptico

$$N = \frac{K(k)K(\sqrt{1-k_1^2})}{K(k_1)K(\sqrt{1-k^2})} \quad k = \frac{\Omega_p}{\Omega_s}, \quad k_1 = \frac{\epsilon}{\sqrt{A^2-1}} \quad K(x) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1-x^2 \sin^2 \theta}}$$

```
function [b,a] = u_elipap(N,Rp,As,Omegac);  
% Unnormalized Elliptic Analog Lowpass Filter Prototype  
% -----  
% [b,a] = u_elipap(N,Rp,As,Omegac);  
%     b = numerator polynomial coefficients  
%     a = denominator polynomial coefficients  
%     N = Order of the Elliptic Filter  
%     Rp = Passband Ripple in dB; Rp > 0  
%     As = Stopband Attenuation in dB; As > 0  
% Omegac = Cutoff frequency in radians/sec  
%  
[z,p,k] = ellipap(N,Rp,As);  
a = real(poly(p)); aNn = a(N+1);  
p = p*Omegac; a = real(poly(p)); aNu = a(N+1);  
b = real(poly(z)); M = length(b); bNn = b(M);  
z = z*Omegac; b = real(poly(z)); bNu = b(M);  
k = k*(aNu*bNn)/(aNn*bNu);  
b0 = k; b = k*b;
```

CARACTERÍSTICAS DE FILTROS ANALÓGICOS PROTÓTIPOS

- Filtro Elíptico

```
function [b,a] = afd_elip(Wp,Ws,Rp,As);  
% Analog Lowpass Filter Design: Elliptic  
% -----  
% [b,a] = afd_elip(Wp,Ws,Rp,As);  
% b = Numerator coefficients of Ha(s)  
% a = Denominator coefficients of Ha(s)  
% Wp = Passband edge frequency in rad/sec; Wp > 0  
% Ws = Stopband edge frequency in rad/sec; Ws > Wp > 0  
% Rp = Passband ripple in +dB; (Rp > 0)  
% As = Stopband attenuation in +dB; (As > 0)  
%  
if Wp <= 0  
    error('Passband edge must be larger than 0')  
end  
if Ws <= Wp  
    error('Stopband edge must be larger than Passband edge')  
end  
if (Rp <= 0) | (As < 0)  
    error('PB ripple and/or SB attenuation ust be larger than 0')  
end  
  
ep = sqrt(10^(Rp/10)-1);  A = 10^(As/20);  
OmegaC = Wp;  k = Wp/Ws;  k1 = ep/sqrt(A*A-1);
```

- **EXAMPLE 8.8** Design an analog elliptic lowpass filter to satisfy the following specifications of Example 8.5:

$$\Omega_p = 0.2\pi, R_p = 1 \text{ dB}$$

$$\Omega_s = 0.3\pi, A_s = 16 \text{ dB}$$

Solution

MATLAB script:

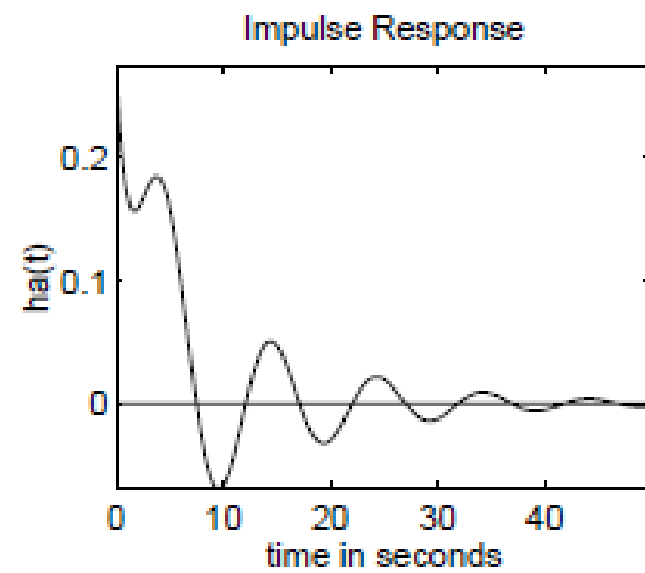
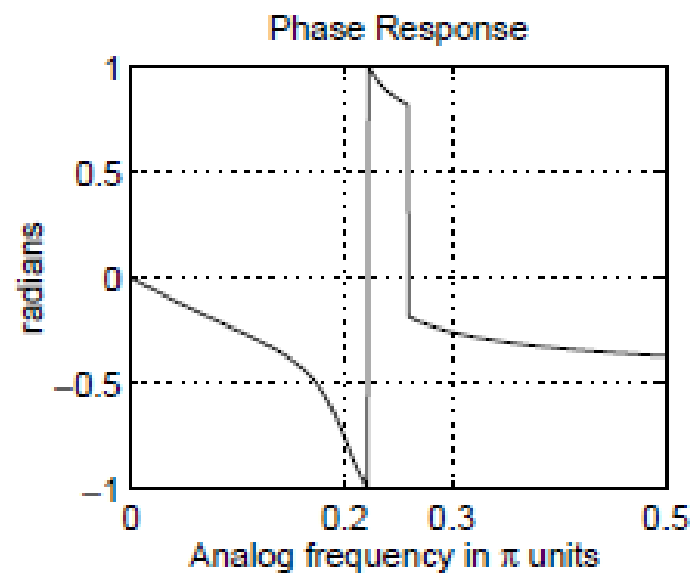
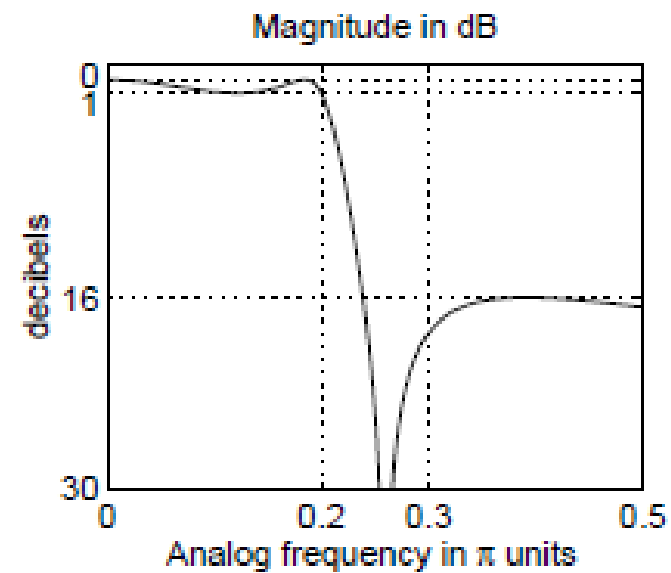
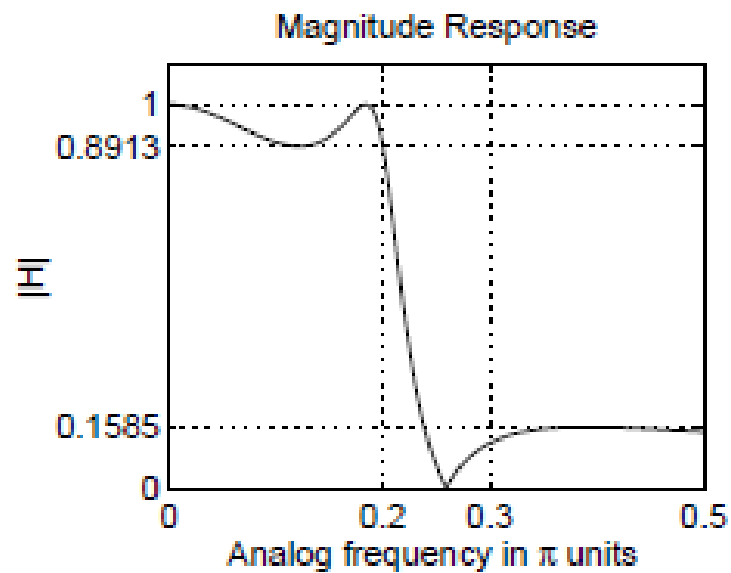
```
>> Wp = 0.2*pi; Ws = 0.3*pi; Rp = 1; As = 16;
>> Ripple = 10 ^ (-Rp/20); Attn = 10 ^ (-As/20);
>> % Analog filter design:
>> [b,a] = afd_elip(Wp,Ws,Rp,As);
*** Elliptic Filter Order = 3
>> % Calculation of second-order sections:
>> [C,B,A] = sdir2cas(b,a)
C = 0.2740
B = 1.0000      0      0.6641
A = 1.0000      0.1696      0.4102
      0      1.0000      0.4435
>> % Calculation of Frequency Response:
>> [db,mag,pha,w] = freqs_m(b,a,0.5*pi);
>> % Calculation of Impulse response:
>> [ha,x,t] = impulse(b,a);
```

The specifications are satisfied by a 3rd-order elliptic filter whose system function is

$$H_a(s) = \frac{0.274 (s^2 + 0.6641)}{(s^2 + 0.1696s + 0.4102)(s + 0.4435)}$$

The filter plots are shown in Figure 8.18.

□



TRANSFORMAÇÃO DE FILTROS ANALÓGICOS PARA DIGITAIS

- Transformação por invariância de impulso

$$\omega = \Omega T \text{ or } e^{j\omega} = e^{j\Omega T} \quad z = e^{j\omega} \quad s = j\Omega \quad z = e^{sT}$$

8.4.2 DESIGN PROCEDURE

Given the digital lowpass filter specifications ω_p , ω_s , R_p , and A_s , we want to determine $H(z)$ by first designing an equivalent analog filter and then mapping it into the desired digital filter. The steps required for this procedure are

1. Choose T and determine the analog frequencies

$$\Omega_p = \frac{\omega_p}{T_p} \quad \text{and} \quad \Omega_s = \frac{\omega_s}{T}$$

2. Design an analog filter $H_a(s)$ using the specifications Ω_p , Ω_s , R_p , and A_s . This can be done using any one of the three (Butterworth, Chebyshev, or elliptic) prototypes of the previous section.
3. Using partial fraction expansion, expand $H_a(s)$ into

$$H_a(s) = \sum_{k=1}^N \frac{R_k}{s - p_k}$$

4. Now transform analog poles $\{p_k\}$ into digital poles $\{e^{p_k T}\}$ to obtain the digital filter:

$$H(z) = \sum_{k=1}^N \frac{R_k}{1 - e^{p_k T} z^{-1}} \quad (8.64)$$

TRANSFORMAÇÃO DE FILTROS ANALÓGICOS PARA DIGITAIS

- Transformação por invariância de impulso

□ EXAMPLE 8.9 Transform

$$H_a(s) = \frac{s+1}{s^2+5s+6}$$

into a digital filter $H(z)$ using the impulse invariance technique in which $T = 0.1$.

Solution

We first expand $H_a(s)$ using partial fraction expansion:

$$H_a(s) = \frac{s+1}{s^2+5s+6} = \frac{2}{s+3} - \frac{1}{s+2}$$

The poles are at $p_1 = -3$ and $p_2 = -2$. Then from (8.64) and using $T = 0.1$, we obtain

$$H(z) = \frac{2}{1 - e^{-3T}z^{-1}} - \frac{1}{1 - e^{-2T}z^{-1}} = \frac{1 - 0.8966z^{-1}}{1 - 1.5595z^{-1} + 0.6065z^{-2}}$$

TRANSFORMAÇÃO DE FILTROS ANALÓGICOS PARA DIGITAIS

- Transformação por invariância de impulso

```
function [b,a] = imp_invr(c,d,T)
% Impulse Invariance Transformation from Analog to Digital Filter
% -----
% [b,a] = imp_invr(c,d,T)
% b = Numerator polynomial in z(-1) of the digital filter
% a = Denominator polynomial in z(-1) of the digital filter
% c = Numerator polynomial in s of the analog filter
% d = Denominator polynomial in s of the analog filter
% T = Sampling (transformation) parameter
%
[R,p,k] = residue(c,d); p = exp(p*T);
[b,a] = residuez(R,p,k); b = real(b'); a = real(a');
```

TRANSFORMAÇÃO DE FILTROS ANALÓGICOS PARA DIGITAIS

- Transformação por invariância de impulso

□ **EXAMPLE 8.10** We demonstrate the use of the `imp_invr` function on the system function from Example 8.9.

Solution

MATLAB script:

```
>> c = [1,1]; d = [1,5,6]; T = 0.1;  
>> [b,a] = imp_invr(c,d,T)  
b = 1.0000 -0.8966  
a = 1.0000 -1.5595 0.6065
```

The digital filter is

$$H(z) = \frac{1 - 0.8966z^{-1}}{1 - 1.5595z^{-1} + 0.6065z^{-2}}$$

□ **EXAMPLE 8.11** Design a lowpass digital filter using a Butterworth prototype to satisfy

$$\omega_p = 0.2\pi, R_p = 1 \text{ dB}$$

$$\omega_s = 0.3\pi, A_s = 15 \text{ dB}$$

Solution

The design procedure is described in the following MATLAB script:

```
>> % Digital Filter Specifications:
>> wp = 0.2*pi;                % digital Passband freq in Hz
>> ws = 0.3*pi;                % digital Stopband freq in Hz
>> Rp = 1;                     % Passband ripple in dB
>> As = 15;                    % Stopband attenuation in dB
>> % Analog Prototype Specifications: Inverse mapping for frequencies
>> T = 1;                      % Set T=1
>> OmegaP = wp / T;            % Prototype Passband freq
>> OmegaS = ws / T;            % Prototype Stopband freq

>> % Analog Butterworth Prototype Filter Calculation:
>> [cs,ds] = afd_butt(OmegaP,OmegaS,Rp,As);
*** Butterworth Filter Order = 6

>> % Impulse Invariance transformation:
>> [b,a] = imp_invr(cs,ds,T); [C,B,A] = dir2par(b,a)
C = []
B = 1.8557    -0.6304
    -2.1428    1.1454
    0.2871    -0.4466
A = 1.0000    -0.9973    0.2570
    1.0000    -1.0691    0.3699
    1.0000    -1.2972    0.6949
```

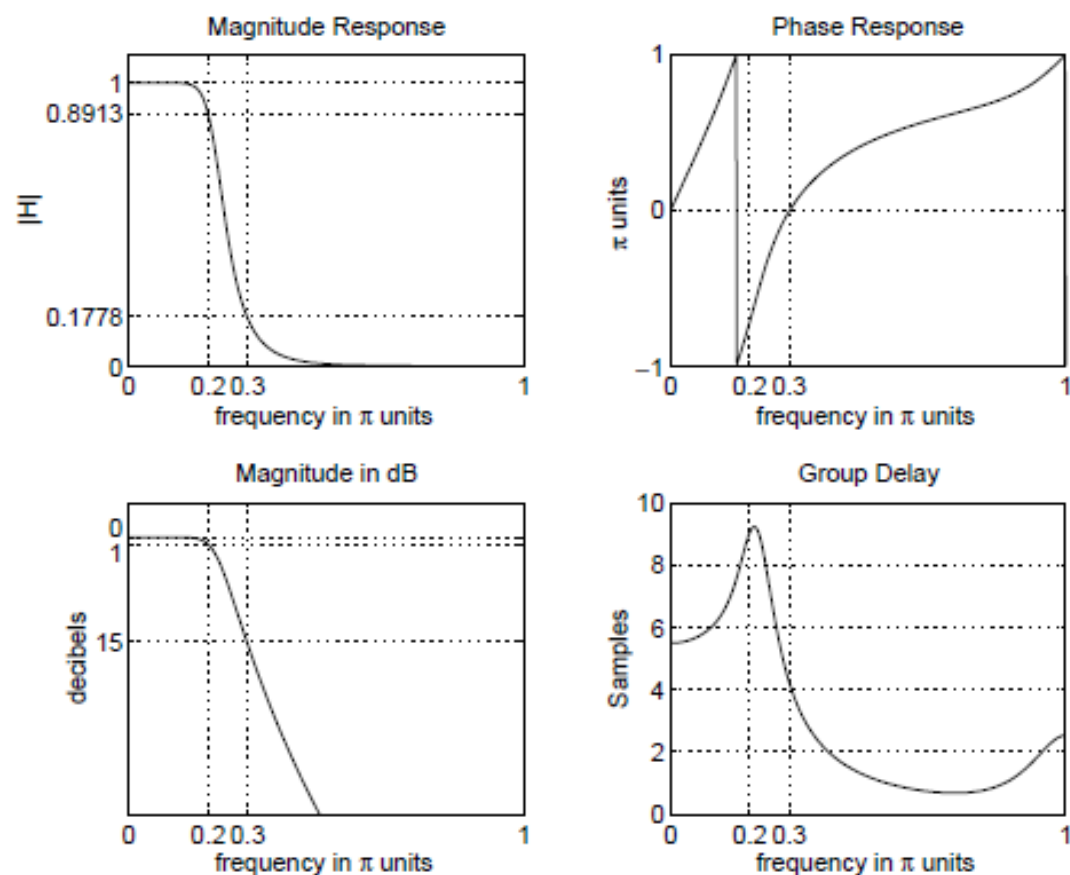


FIGURE 8.21 *Digital Butterworth lowpass filter using impulse invariance design*

The desired filter is a 6th-order Butterworth filter whose system function $H(z)$ is given in the parallel form

$$H(z) = \frac{1.8587 - 0.6304z^{-1}}{1 - 0.9973z^{-1} + 0.257z^{-2}} + \frac{-2.1428 + 1.1454z^{-1}}{1 - 1.0691z^{-1} + 0.3699z^{-2}} + \frac{0.2871 - 0.4463z^{-1}}{1 - 1.2972z^{-1} + 0.6449z^{-2}}$$

□ **EXAMPLE 8.12** Design a lowpass digital filter using a Chebyshev-I prototype to satisfy

$$\omega_p = 0.2\pi, R_p = 1 \text{ dB}$$

$$\omega_s = 0.3\pi, A_s = 15 \text{ dB}$$

```
>> % Digital Filter Specifications:
>> wp = 0.2*pi;           % digital Passband freq in rad
>> ws = 0.3*pi;           % digital Stopband freq in rad
>> Rp = 1;                 % Passband ripple in dB
>> As = 15;                % Stopband attenuation in dB

>> % Analog Prototype Specifications: Inverse mapping for frequencies
>> T = 1;                  % Set T=1
>> OmegaP = wp / T;        % Prototype Passband freq
>> OmegaS = ws / T;        % Prototype Stopband freq
>> % Analog Chebyshev-1 Prototype Filter Calculation:
>> [cs,ds] = afd_chb1(OmegaP,OmegaS,Rp,As);
*** Chebyshev-1 Filter Order = 4

>> % Impulse Invariance transformation:
>> [b,a] = imp_invr(cs,ds,T); [C,B,A] = dir2par(b,a)
C = []
B = -0.0833    -0.0246
      0.0833     0.0239
A = 1.0000    -1.4934     0.8392
      1.0000    -1.5658     0.6549
```

The desired filter is a 4th-order Chebyshev-I filter whose system function $H(z)$ is

$$H(z) = \frac{-0.0833 - 0.0246z^{-1}}{1 - 1.4934z^{-1} + 0.8392z^{-2}} + \frac{-0.0833 + 0.0239z^{-1}}{1 - 1.5658z^{-1} + 0.6549z^{-2}}$$

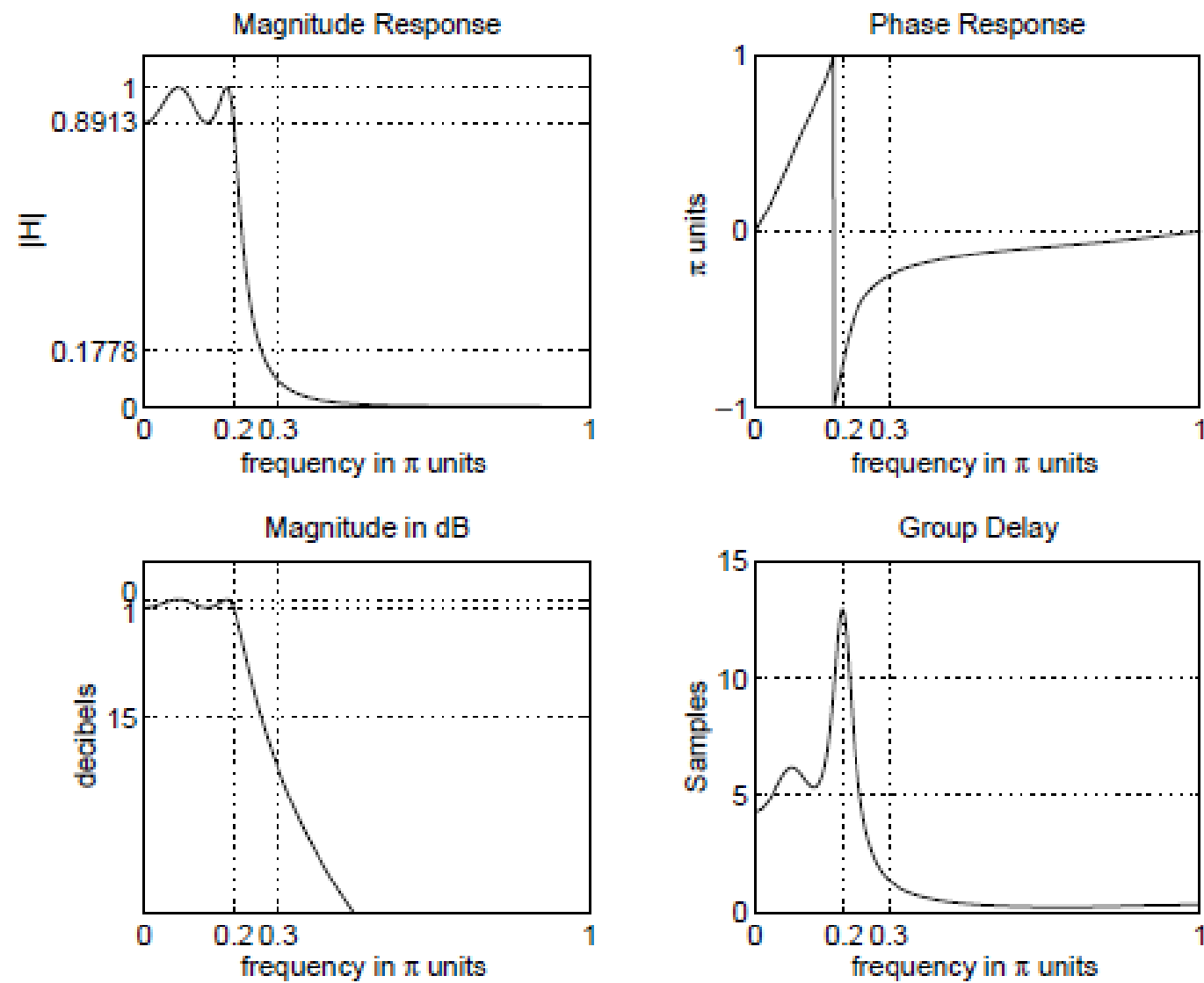


FIGURE 8.22 *Digital Chebyshev-I lowpass filter using impulse invariance design*

□ **EXAMPLE 8.13** Design a lowpass digital filter using a Chebyshev-II prototype to satisfy

$$\omega_p = 0.2\pi, R_p = 1 \text{ dB}$$

$$\omega_s = 0.3\pi, A_s = 15 \text{ dB}$$

```
>> % Digital Filter Specifications:
>> wp = 0.2*pi;           % digital Passband freq in rad
>> ws = 0.3*pi;           % digital Stopband freq in rad
>> Rp = 1;                % Passband ripple in dB
>> As = 15;               % Stopband attenuation in dB

>> % Analog Prototype Specifications: Inverse mapping for frequencies
>> T = 1;                 % Set T=1
>> OmegaP = wp / T;       % Prototype Passband freq
>> OmegaS = ws / T;       % Prototype Stopband freq

>> % Analog Chebyshev-1 Prototype Filter Calculation:
>> [cs,ds] = afd_chb2(OmegaP,OmegaS,Rp,As);
*** Chebyshev-2 Filter Order = 4

>> % Impulse Invariance transformation:
>> [b,a] = imp_invr(cs,ds,T); [C,B,A] = dir2par(b,a);
```

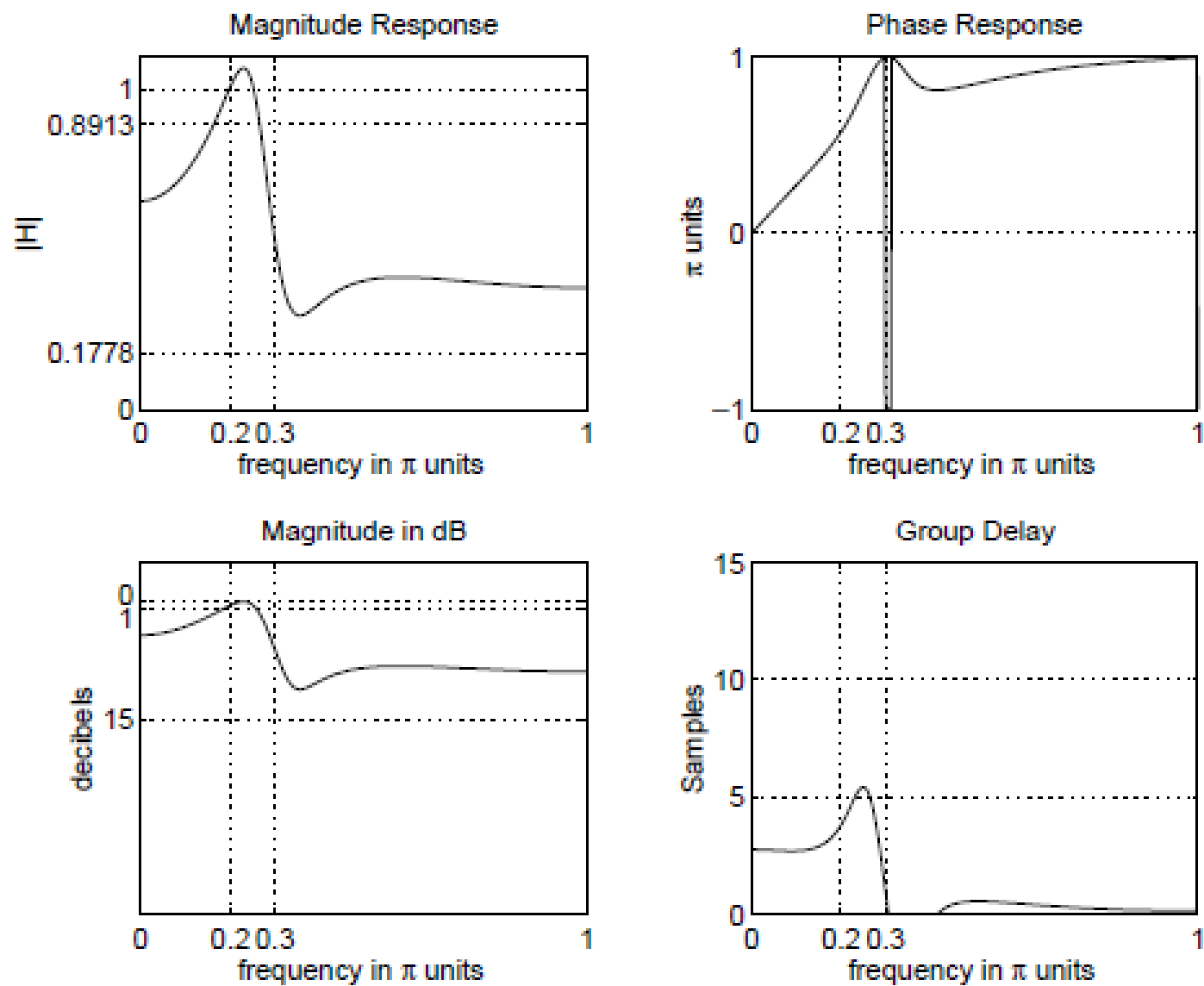


FIGURE 8.23 *Digital Chebyshev-II lowpass filter using impulse invariance design*

□ **EXAMPLE 8.14** Design a lowpass digital filter using an elliptic prototype to satisfy

$$\omega_p = 0.2\pi, R_p = 1 \text{ dB}$$

$$\omega_s = 0.3\pi, A_s = 15 \text{ dB}$$

Solution

The elliptic filter is equiripple in both bands. Hence this situation is similar to that of the Chebyshev-II filter, and we should not expect a good digital filter. The MATLAB script follows:

```
>> % Digital Filter Specifications:
>> wp = 0.2*pi;           % digital Passband freq in rad
>> ws = 0.3*pi;           % digital Stopband freq in rad
>> Rp = 1;                % Passband ripple in dB
>> As = 15;               % Stopband attenuation in dB

>> % Analog Prototype Specifications: Inverse mapping for frequencies
>> T = 1;                 % Set T=1
>> OmegaP = wp / T;       % Prototype Passband freq
>> OmegaS = ws / T;       % Prototype Stopband freq

>> % Analog Elliptic Prototype Filter Calculation:
>> [cs,ds] = afd_elip(OmegaP,OmegaS,Rp,As);
*** Elliptic Filter Order = 3

>> % Impulse Invariance transformation:
>> [b,a] = imp_invr(cs,ds,T); [C,B,A] = dir2par(b,a);
```

TRANSFORMAÇÃO DE FILTROS ANALÓGICOS PARA DIGITAIS

- Transformação Bilinear

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}} \implies z = \frac{1 + sT/2}{1 - sT/2}$$

□ **EXAMPLE 8.15** Transform $H_a(s) = \frac{s + 1}{s^2 + 5s + 6}$ into a digital filter using the bilinear transformation. Choose $T = 1$.

Solution

Using (8.65), we obtain

$$\begin{aligned} H(z) &= H_a \left(\left. \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}} \right|_{T=1} \right) = H_a \left(2 \frac{1 - z^{-1}}{1 + z^{-1}} \right) \\ &= \frac{2 \frac{1 - z^{-1}}{1 + z^{-1}} + 1}{\left(2 \frac{1 - z^{-1}}{1 + z^{-1}} \right)^2 + 5 \left(2 \frac{1 - z^{-1}}{1 + z^{-1}} \right) + 6} \end{aligned}$$

Simplifying,

$$H(z) = \frac{3 + 2z^{-1} - z^{-2}}{20 + 4z^{-1}} = \frac{0.15 + 0.1z^{-1} - 0.05z^{-2}}{1 + 0.2z^{-1}}$$

□ **EXAMPLE 8.16** Transform the system function $H_a(s)$ in Example 8.15 using the **bilinear** function.

Solution

MATLAB script:

```
>> c = [1,1]; d = [1,5,6]; T = 1; Fs = 1/T;  
>> [b,a] = bilinear(c,d,Fs)  
b = 0.1500    0.1000   -0.0500  
a = 1.0000    0.2000    0.0000
```

The filter is

$$H(z) = \frac{0.15 + 0.1z^{-1} - 0.05z^{-2}}{1 + 0.2z^{-1}}$$

as before.

□

8.4.4 DESIGN PROCEDURE

Given digital filter specifications ω_p , ω_s , R_p , and A_s , we want to determine $H(z)$. The design steps in this procedure are the following:

1. Choose a value for T . This is arbitrary, and we may set $T = 1$.
2. Prewarp the cutoff frequencies ω_p and ω_s ; that is, calculate Ω_p and Ω_s using (8.67):

$$\Omega_p = \frac{2}{T} \tan\left(\frac{\omega_p}{2}\right), \quad \Omega_s = \frac{2}{T} \tan\left(\frac{\omega_s}{2}\right) \quad (8.68)$$

3. Design an analog filter $H_a(s)$ to meet the specifications Ω_p , Ω_s , R_p , and A_s . We have already described how to do this in the previous section.
4. Finally, set

$$H(z) = H_a\left(\frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}\right)$$

and simplify to obtain $H(z)$ as a rational function in z^{-1} .

In the next several examples we demonstrate this design procedure on our analog prototype filters.

□ **EXAMPLE 8.17** Design the digital Butterworth filter of Example 8.11. The specifications are

$$\omega_p = 0.2\pi, R_p = 1 \text{ dB}$$

$$\omega_s = 0.3\pi, A_s = 15 \text{ dB}$$

```
>> % Digital Filter Specifications:
>> wp = 0.2*pi;                % digital Passband freq in rad
>> ws = 0.3*pi;                % digital Stopband freq in rad
>> Rp = 1;                     % Passband ripple in dB
>> As = 15;                    % Stopband attenuation in dB
>> % Analog Prototype Specifications: Inverse mapping for frequencies
>> T = 1; Fs = 1/T;            % Set T=1
>> OmegaP = (2/T)*tan(wp/2);    % Prewarp Prototype Passband freq
>> OmegaS = (2/T)*tan(ws/2);    % Prewarp Prototype Stopband freq
>> % Analog Butterworth Prototype Filter Calculation:
>> [cs,ds] = afd_butt(OmegaP,OmegaS,Rp,As);
*** Butterworth Filter Order = 6
>> % Bilinear transformation:
>> [b,a] = bilinear(cs,ds,Fs); [C,B,A] = dir2cas(b,a)
C = 5.7969e-004
B = 1.0000    2.0183    1.0186
    1.0000    1.9814    0.9817
    1.0000    2.0004    1.0000
A = 1.0000   -0.9459    0.2342
    1.0000   -1.0541    0.3753
    1.0000   -1.3143    0.7149
```

TRANSFORMAÇÃO DE FREQUENCIA

TABLE 8.2 Frequency transformation for digital filters (prototype lowpass filter has cutoff frequency ω'_c)

Type of Transformation	Transformation	Parameters
Lowpass	$z^{-1} \longrightarrow \frac{z^{-1} - \alpha}{1 - \alpha z^{-1}}$	ω_c = cutoff frequency of new filter $\alpha = \frac{\sin [(\omega'_c - \omega_c) / 2]}{\sin [(\omega'_c + \omega_c) / 2]}$
Highpass	$z^{-1} \longrightarrow -\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}}$	ω_c = cutoff frequency of new filter $\alpha = -\frac{\cos [(\omega'_c + \omega_c) / 2]}{\cos [(\omega'_c - \omega_c) / 2]}$
Bandpass	$z^{-1} \longrightarrow -\frac{z^{-2} - \alpha_1 z^{-1} + \alpha_2}{\alpha_2 z^{-2} - \alpha_1 z^{-1} + 1}$	ω_ℓ = lower cutoff frequency ω_u = upper cutoff frequency $\alpha_1 = -2\beta K / (K + 1)$ $\alpha_2 = (K - 1) / (K + 1)$ $\beta = \frac{\cos [(\omega_u + \omega_\ell) / 2]}{\cos [(\omega_u - \omega_\ell) / 2]}$ $K = \cot \frac{\omega_u - \omega_\ell}{2} \tan \frac{\omega'_c}{2}$
Bandstop	$z^{-1} \longrightarrow \frac{z^{-2} - \alpha_1 z^{-1} + \alpha_2}{\alpha_2 z^{-2} - \alpha_1 z^{-1} + 1}$	ω_ℓ = lower cutoff frequency ω_u = upper cutoff frequency $\alpha_1 = -2\beta / (K + 1)$ $\alpha_2 = (K - 1) / (K + 1)$ $\beta = \frac{\cos [(\omega_u + \omega_\ell) / 2]}{\cos [(\omega_u - \omega_\ell) / 2]}$ $K = \tan \frac{\omega_u - \omega_\ell}{2} \tan \frac{\omega'_c}{2}$